

Unconstrained Vision Guided UAV Based Safe Helicopter Landing

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Abstract—In this paper, we have addressed the problem of automated detection of safe zone(s) for helicopter landing in hazardous environments from videos captured by an Unmanned Aerial Vehicle (UAV). The unconstrained motion of the video capturing drone (the UAV in our case) makes the problem further difficult. The solution pipeline consists of natural landmark detection and tracking, stereo-pair generation using constrained graph clustering, digital terrain map construction and safe landing zone detection. The main methodological contribution lies in mathematically formulating epipolar constraint and then using it in a Minimum Spanning Tree (MST) based graph clustering approach. We have also made publicly available AHL (Autonomous Helicopter Landing) dataset, a new aerial video dataset captured by a drone, with annotated ground-truths. Experimental comparisons with other competing clustering methods i) in terms of Dunn Index and Davies Bouldin Index as well as ii) for frame-level safe zone detection in terms of F-measure and confusion matrix clearly demonstrate the effectiveness of the proposed formulation.

Index Terms—Vision-guided landing, epipolar constraint, graph clustering, aerial video dataset.

I. INTRODUCTION

With recent technological advancements, demand for computer vision based autonomous systems [1]–[3] like self-driving cars [3], vision-guided Robots (VGRs) [2] have increased manifold. Vision guided autonomous helicopter control [1], [4] has also likewise become an important topic of research. Although works on active sensors based autonomous helicopter landing have already been reported [1], such systems are found to be bulky, expensive, power hungry and often require a specially trained expertise to operate. In sharp contrast cameras are simple, cheap, small and less power consuming. Moreover, visual sensors are rich data source which can lead to better navigation and close proximity flights with relevant feedback from robust image processing algorithms.

Although vision based autonomous helicopter seems promising, the bulky nature [5] of aircraft system makes it unsuitable for emergency tasks like monitoring natural disasters and searching of safe zones for landing. It is also not economical in terms of fuel consumption and less accessible in constricted areas. Unmanned Aerial Vehicles (UAVs) [6], [7] (also known as drones) on the other hand alleviate these challenges thereby increasing the chance of realizing the above tasks. Nowadays, UAVs are equipped with high resolution

visual sensors that can provide detailed imaging and more accurate 3D mapping for monitoring large-scale disaster prone areas such as resulting from flooding and earthquakes. They can be easily deployed to facilitate faster alerts and timely coordination of relief efforts. They can also provide real-time assistance towards safe landing of an aircraft system in disaster/emergency situations.

The goal of this work is to identify safe zones for helicopter landing for possible relief distribution in disaster stricken areas from aerial videos captured by UAVs in an unconstrained manner. Note that avoiding hazardous regions is an important part of autonomous helicopter landing [1]. In normal situations, well-defined landmarks, e.g., a helipad can be easily located using standard image processing techniques [5], [8] and autonomous landing is somewhat less challenging. However, in case of disaster affected regions, such landmarks may not be available at all and this can limit the utility of existing vision based approaches for autonomous landing. While often the problem of detecting and avoiding steep slopes, rock or natural obstacles based on natural landmarks [9] have been studied separately, we have extracted key-points from these natural landmarks for locating safe landing zone. In particular, Speeded-Up Robust Features (SURF) [10] for key-point extraction which is not only faster in computation but also invariant to image translation, rotation and scaling, is used in this work. Additionally, these features are also found to be partially invariant to affine 3D projection and illumination changes.

To properly locate the safe landing zone(s), a robust terrain analysis is necessary which is usually performed over a 3D Digital Elevation Map (DEM) [11]. A good DEM will accurately represent the elevation of each pixel of a scene image. A stereo pair is required to construct a DEM which can come from either a monocular (single camera) [12] or binocular stereo vision. As most available drones are equipped with single cameras we propose a monocular stereo-pair generation approach. Earlier Johnson *et al.* [1] have treated two consecutive frames as stereo pairs for DEM construction for autonomous helicopter landing. Unlike an helicopter, a drone can exhibit a complete random motion [13] in any arbitrary direction which is likely to violate the epipolar constraint [14] between consecutive frames. Thus, to generate good stereo pairs we propose a Minimum Spanning Tree

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(MST) [15] based constrained graph clustering approach that satisfies epipolar geometry between any two video frames (not necessarily consecutive). The detailed description of the proposed formulation is described in section III.

We have carefully noticed that although the problem of vision guided autonomous helicopter landing is an important topic of research, the area remained relatively unexplored due to unavailability of dataset and flight restriction in urban areas. We are first to propose and render a real aerial dataset. We name our dataset as Autonomous Helicopter Landing (AHL) dataset and have made it publicly available at the link <https://sites.google.com/site/ivprgroup/helicopter-landing>. The dataset is in form of a sequence of frames taken as snapshots during drone flights embedded with cameras. Other relevant information like camera setting and GPS position (latitude, longitude and altitude of drone) are also included. The dataset consists of diverse scenes of both (safe) landing and no-landing zones. We have also provided a ground truth annotation of safe landing zones in form of foreground masks of safe landing regions. A more detailed description is given in section V. We strongly believe that the availability of this dataset would significantly increase the research activity in this area.

In this paper, we have addressed the problem of autonomous safe helicopter landing facilitated by vision guided UAV/drone based system. Our contributions are now summarized below:

- 1) We propose a constrained clustering approach to extract appropriate stereo pairs from a single camera under unrestricted/random trajectory of drone flight path. The constraint is based on epipolar geometry between a pair of video frames. This pair-wise constraint is used to prune the edges to sparsify a fully connected graph constructed over the video frames. Finally, a MST based graph clustering is applied on the sparse weighted graph with frames (as vertices) mutually satisfying the stereo (epipolar) constraint and edge weights being derived from frame similarities.
- 2) We introduce a publicly available AHL dataset containing several scenes of landing and no-landing regions. The dataset is collected from specialized drone based system in real world scenarios with embedded GPS and camera information. We have additionally provided ground truth annotations as foreground masks for frames with safe-landing zones.

We have further shown that our proposed algorithm can be easily deployed and integrated into a drone based embedded system for real-time execution.

II. RELATED WORKS

Vision based autonomous control [7], [8], [16], [17] of aerial vehicles have been an active research area for some-time. Earlier, Benedetti and Perona [16] have utilized motion estimates from video frames along with GPS and sonar based altimeter measurements for helicopter navigation. Amidi *et al.* [4] developed a visual odometer to estimate helicopter motion and location by locking on to target ground feature. In [8], the authors designed a vision based followup system

to track and locate a known area for successful landing. The authors in [9] proposed a safe-landing site detection which is segmented from an unstructured terrain. In the above methods, the safe landing location is known a priori and identification of safe zone is not really required. This makes the problem relatively simple as compared to ours.

Now, we discuss a set of methods focused on safe zone detection for landing. For example, Ghamisi *et al.* [18] indicated possible safe landing zones based on color and texture features. However, a flat region as a safe zone cannot be always rightly represented with a specific color or texture [7]. A similar work is done by Huang *et al.* [6], where, they employed shape based level-set with fuzzy c-means clustering to segment dominant color regions in a scene. The method works well if the dominant region mostly consists of flat surfaces. Hinzmann *et al.* [19] proposed a shape based approach by combining texture with scene geometry for proper landing site detection with a prior knowledge of the environment. However, natural landing site can vary in shape or size and thus assumption of any specific shape limits the applicability of such approach. Patterson *et al.* [17] employed Canny edge feature along with a fuzzy approach for region-of-interest classification for timely safe-zone identification for UAV. They have separated regions with trees, paths and rivers and labeled other homogeneous regions as safe zones. However, the method perform unsatisfactorily in complex scenarios due to sensitivity of the Canny edge detector to noise. In a recent method as proposed by Kaliahi *et al.* [7] for automatic safe zone detection of UAVs, Gabor Transform was first applied to obtain a set of different Gabor-oriented images. This is followed by histogram thresholding operation and connected component analysis for extraction of candidate pixels (CPs). Markov Chain Codes (MCCs) are finally generated which determine the probability of candidate pixels belongs to belong to a single region. The above review indicates that most methods on safe landing zone detection are based on color, edge, texture and other simple geometric features which may not be robust enough to be applicable for complex landing regions. Most method on safe landing regions are actually for UAV landing rather than helicopter landing. Our proposed method does not rely on simple features, and can easily adapt to arbitrary/unconstrained change of scenes that a drone may encounter during its flight.

The work closest to this paper is the one reported by Johnson *et al.* [1]. However, still there are many differences between the two approaches. The work described in [1] is based on constrained image capturing as the whole vision system is mounted and conducted from the helicopter itself. In sharp contrast, our method being UAV driven, we have to deal with unconstrained motion of the drone. So, an epipolar constraint guided MST based graph clustering algorithm is proposed in the paper. Secondly, we have computed Digital elevation map (DEM) using robust 3D Delaunay triangulation while the authors in [1] have used a simple bilinear interpolation to model an uneven surface. Thirdly, our slope and roughness map computations are based on robust surface fitting in contrast to piecewise fitting by Johnson *et al.*

III. PROPOSED FRAMEWORK

In this section, we provide a detailed description of the proposed framework. We have followed the overall baseline framework of Johnson *et al.* [1] with differences as stated in the last paragraph. Our solution pipeline consists of four major steps, namely, natural landmark detection and tracking, stereo-pair generation using constrained graph clustering, digital terrain map construction and safe landing zone detection. Each of the above steps are now described below.

A. Natural Landmark Detection and Tracking

Owing to the repeatable nature of natural landmarks [20], they together form an efficient way to locate a desired region-of-interest (ROI). We have utilized robust SURF based key-points for detecting and tracking natural landmarks. We have also built a frame representation based on these detected key-points for robust stereo-pair generation at a later stage (see section III-B1) of our pipeline.

1) **Robust Key-point Detection and Tracking:** SURF uses Hessian matrix to detect interest points. For any point $p(x, y)$ in an image $\mathbf{I}$, the hessian matrix $\mathbf{H}(p, \sigma)$ at scale σ is defined as:

$$\mathbf{H}(p, \sigma) = \begin{bmatrix} L_{xx}(p, \sigma) & L_{xy}(p, \sigma) \\ L_{yx}(p, \sigma) & L_{yy}(p, \sigma) \end{bmatrix} \quad (1)$$

where $L_{xx}(p, \sigma)$, $L_{xy}(p, \sigma)$, $L_{yx}(p, \sigma)$ and $L_{yy}(p, \sigma)$ are convolutions of an image $\mathbf{I}(x, y)$ (a video frame in this case) with Gaussian second order derivatives $\frac{\partial^2}{\partial x^2}g(p, \sigma)$, $\frac{\partial^2}{\partial x \partial y}g(p, \sigma)$, $\frac{\partial^2}{\partial y \partial x}g(p, \sigma)$ and $\frac{\partial^2}{\partial y^2}g(p, \sigma)$ at point $p(x, y)$ respectively. To speed up the computation, second order Gaussian derivatives in the Hessian matrix are approximated using box filters. To detect key-point at different scales, a scale space representation is obtained by varying σ followed by a non-maximal suppression over the scales. Each region around $p(x, y)$ is split into different sub-regions. Finally, the SURF feature is constructed by concatenating all the feature vectors V_k for the k^{th} sub-region, i.e., $\mathbf{v}_k = [\sum_k dx, \sum_k dy, \sum_k |dx|, \sum_k |dy|]$, where, dx and dy are Haar wavelet responses [10] along x and y direction. Key-points between two frames are tracked by matching features using a fast approximate nearest neighbor search algorithm [21].

2) **Frame representation using SURF:** For performing a graph based clustering (see section III-B2) over video frames we need to construct a single feature representation for each frame from several detected key-points. Number of detected key-points varies with frame and each key-point is having 64-dimensional feature vector (16 sub-regions each with 4 elements of $\mathbf{v}_k$). Thus, for a frame i , we have taken the mean feature vector for the k^{th} sub-region as $\mathbf{v}_k = \frac{1}{N} [\sum_n \sum_k dx, \sum_n \sum_k dy, \sum_n \sum_k |dx|, \sum_n \sum_k |dy|]$ over $n \in [1, 2, \dots, N]$ extracted key-points. The final feature vector for frame is $\mathbf{f}_i = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \dots, \mathbf{v}_{16}]$. Geometrically, the feature vector $\mathbf{f}_i$ represents the centroid of N extracted key-points in a 64-dimensional feature space.

B. Stereo-Pair Generation using Constrained Clustering

Constructing a 3D digital surface model [11] of a real world generally requires a binocular stereo pair. However, several works [22] have demonstrated that developing a monocular (single camera) stereo system is possible. Structure from motion [12] is one such approach that uses 2D image sequence to generate a 3D structure. Such an approach is applicable to small local changes in a scene and gives erroneous result with rapid scene changes (due to randomized UAV motion). To handle such rapid changes we have formulated a constrained graph clustering approach for successful determination of stereo pairs from an unconstrained video capturing by a drone with random motion.

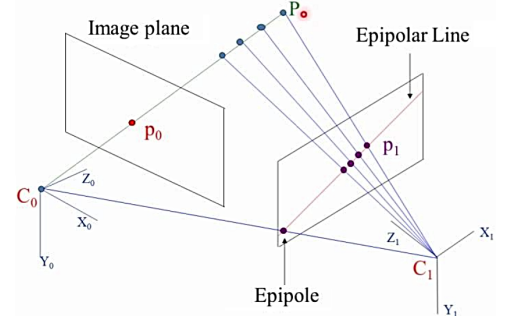


Fig. 1. Epipolar geometry between two frames

1) **Formulation of Epipolar Constraint:** Two image planes captured from well calibrated cameras will form a stereo pair if they satisfy the epipolar constraint [14]. This is equivalent to saying the three vectors in figure 1, namely, $\overrightarrow{C_0P_0}$, $\overrightarrow{C_0C_1}$ and $\overrightarrow{C_1P_1}$ must be coplanar. So, we write

$$\overrightarrow{C_0P_0} \cdot [\overrightarrow{C_0C_1} \times \overrightarrow{C_1P_1}] = 0 \quad (2)$$

Let us write $\mathbf{p}' = \overrightarrow{C_0P_0}$ and $\mathbf{p} = \overrightarrow{C_1P_1}$. Then equation 2 can be re-written independent of coordinates as:

$$\begin{aligned} \mathbf{p}' \cdot [\mathbf{t} \times (\mathcal{R}\mathbf{p})] &= 0 \\ \implies \mathbf{p}'^T \mathcal{E} \mathbf{p} &= 0 \end{aligned} \quad (3)$$

where, $\mathbf{t}$ and $\mathcal{R}$ denote translation vector and rotation matrix respectively which together capture the motion of a point from $\overrightarrow{C_0C_1}$ to $\overrightarrow{C_1P_1}$. Here, $\mathcal{E}$ is called essential or fundamental matrix that represent this transformation. Let the set of N matched key-points in an image pair be represented by two ordered sets $\mathbf{P}$ and $\mathbf{P}'$ respectively. So, $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_k, \dots, \mathbf{p}_N]$ and $\mathbf{P}' = [\mathbf{p}'_1, \mathbf{p}'_2, \dots, \mathbf{p}'_k, \dots, \mathbf{p}'_N]$. The k^{th} matched pair $\{\mathbf{p}_k, \mathbf{p}'_k\}$ can be represented by homogeneous coordinate system as, $\mathbf{p}_k = \begin{pmatrix} x_k \\ y_k \\ 1 \end{pmatrix} \in \mathbb{R}^{3 \times 1}$ and $\mathbf{p}'_k = \begin{pmatrix} x'_k \\ y'_k \\ 1 \end{pmatrix} \in \mathbb{R}^{3 \times 1}$. Without loss of generality, any matched key-point pair $\{\mathbf{p}_k, \mathbf{p}'_k\}$ satisfying epipolar constraint (equation 3) can be represented as:

$$\mathbf{p}'_k^T \mathcal{E} \mathbf{p}_k = 0 \quad (4)$$

where, $\mathcal{E} \in \mathbb{R}^{3 \times 3}$. Physically, $\mathcal{E} \mathbf{p}_k$ is the epipolar line where the point $\mathbf{p}'_k$ must lie (see figure 1). Our objective is to estimate the fundamental matrix $\mathcal{E}$ that satisfies the above

condition for all matched key-point pairs. But it is not always guaranteed that all matched pairs can be linked with same epipolar geometry. Therefore, a subset M of N matched key point pairs are selected can nearly satisfy the epipolar constraint. Let us denote them by $\mathbf{Q} = [\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_k, \dots, \mathbf{q}_M]$ and $\mathbf{Q}' = [\mathbf{q}'_1, \mathbf{q}'_2, \dots, \mathbf{q}'_k, \dots, \mathbf{q}'_M]$ such that $\mathbf{Q} \subset \mathbf{P}$ and $\mathbf{Q}' \subset \mathbf{P}'$. So, we modify the previous equation for $\{\mathbf{q}_k, \mathbf{q}'_k\}$ and write:

$$\mathbf{q}'_k{}^T \mathcal{E} \mathbf{q}_k < \epsilon_0 \quad (5)$$

where, ϵ_0 is a constant. Pairs $\{\mathbf{q}'_k, \mathbf{q}_k\}$ are known as epipolar inliers and are selected in a random consensus fashion (RANSAC). An overall fitness value of two given frames to be a stereo pair can be represented at iteration t in form of a matrix operation as:

$$F_{epi} = \frac{1}{M} \text{Tr}(\mathbf{Q}'^T \mathcal{E}^t \mathbf{Q}) \quad (6)$$

Here, $\text{Tr}(\cdot)$ is the trace of a matrix. The fundamental matrix $\mathcal{E}$ is estimated using M-point algorithm while minimizing F_{epi} at iteration t . Thus, based on the above fitness value the epipolar constraint between two frames are written as:

$$\begin{aligned} \frac{1}{M} \text{Tr}(\mathbf{Q}'^T \mathcal{E}^t \mathbf{Q}) &< \epsilon \\ \Rightarrow \text{Tr}(\mathbf{Q}'^T \mathcal{E}^t \mathbf{Q}) &< \epsilon M \\ \Rightarrow \text{Tr}(\mathbf{Q}'^T \mathcal{E}^\tau \mathbf{Q}) &< \delta \end{aligned} \quad (7)$$

where, $\delta = \epsilon M$ is the overall bound for a given image pair and τ is the final iteration. The physical interpretation of imposing such constraint is that two frames can only be in the same group (cluster) if and only if their fitness are bounded by δ .

2) Construction of a Video Representation Graph: A complete weighted graph is initially constructed in the 64-dimensional feature space with all frames in a video segment. We term this as a video representation graph (VRG). Each frame is deemed as a vertex. Edges are established between each pair of vertices. The edge weight w_{ij} between the frames i and j with respective feature vectors $\mathbf{f}_i$ and $\mathbf{f}_j$ is given by:

$$w_{ij} = \sum_{k=1}^{64} |f_{ik} - f_{jk}| \quad (8)$$

We choose the city-block distance for faster computation. Now, the epipolar constraint in equation (7) is used to prune the complete graph to form a sparse graph. So, for edge weights, we write:

$$e_{ij} = \begin{cases} w_{ij} & \text{Tr}(\mathbf{Q}_i^T \mathcal{E}_{ij}^\tau \mathbf{Q}_j) < \delta \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

Here, $\mathbf{Q}_i$ and $\mathbf{Q}_j$ are matrices of matched set of key-point pairs from frame i and j with connecting edge e_{ij} . Typically, a video segment with 230 frames would consists of ${}^{230}C_2 \approx 25000$ edges which reduces to a sparse graph with ≈ 5000 vertices. This sparse VRG is denoted by $G(V, E)$ where E and V are edges and vertices.

3) MST based Sparse Graph Clustering: We apply a MST based clustering on this sparse graph $G(V, E)$. The MST

based clustering algorithm provides some inherent advantages. Firstly, it can also detect irregular shaped clusters including non-convex ones. Secondly, it does not require cluster number in advance. In this clustering approach the inconsistent edges need to be removed. Those edges, whose weights are significantly greater than mean weight of the nearby edges, are deemed as inconsistent. Let e_{ij} denotes an edge in the MST connecting the vertices i and j having weight w_{ij} . Let N_i and N_j be the set of direct neighborhoods of i and j respectively. Following the normal distribution of edge weights in MST [15], the inconsistency measure can be formulated as:

$$w_{ij} > \max(\bar{w}_{N_i} + \sigma_{N_i}, \bar{w}_{N_j} + \sigma_{N_j}) \quad (10)$$

where, $\bar{w}_{N_i}$ and $\bar{w}_{N_j}$ are the average weights of the edges in N_i and N_j respectively. Similarly, σ_{N_i} and σ_{N_j} represent the standard deviations of edge weights in N_i and N_j respectively. All edges of the tree satisfying the above inconsistency measure are most likely to be inter-cluster edges and are thereby removed resulting in a disjoint set of sub-trees each representing a separate cluster. Each cluster consists of frames of same scene where the frames are likely to satisfy the epipolar constraint. We then extract stereo image pairs from multiple such clusters as we intend to extract all possible safe-zones from different scenes.

C. Digital Terrain Map Construction

After obtaining several stereo pairs of a particular scene from one of the clusters as described in section III-B2, we first calibrate the camera to obtain the camera intrinsic parameters $\mathbf{K}$. These parameters are represented by:

$$\mathbf{K} = \begin{bmatrix} \alpha_x & 0 & c_x \\ 0 & \alpha_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (11)$$

where, $\alpha_x = f \cdot m_x$ and $\alpha_y = f \cdot m_y$ are camera focal length f represented in terms of pixels along x and y direction scaled with factors m_x and m_y respectively. Further, c_x and c_y are principal focal point which would ideally be the center of the image. Based on the previously obtained Fundamental matrix $\mathcal{E}$ (by solving equation 5) we can now generate 3D point cloud followed by Digital elevation Map (DEM) as discussed below.

1) 3D-Point Cloud Generation: For the generation of the 3D point cloud we need to first estimate the camera pose between two images in terms of the rotation matrix $[\mathbf{R}]_{3 \times 3}$ and the translation vector $[\mathbf{T}]_{3 \times 1}$ that project a 3D world coordinate into a 2D image point. So, we write:

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \mathbf{K} [\mathbf{R} \ \mathbf{T}] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \mathbf{K} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (12)$$

where (x, y) and (X, Y, Z) are image point and its corresponding world coordinate. The matrix $[\mathbf{R} \ \mathbf{T}]$ is known as camera extrinsic matrix. The overall camera projection matrix $\mathbf{P}$ can be computed as the product of camera intrinsic (see equation 11) and extrinsic matrix as $\mathbf{P} = \mathbf{K}[\mathbf{R} \ \mathbf{T}]$. Here, λ is

the homogeneous scaling factor related to the image scene depth. Now, if the matched coordinate points (x, y) and (x', y') are linked by epipolar geometry using fundamental matrix $\mathcal{E}$ then we can write,

$$[\mathbf{R} \ \mathbf{T}]^{-1} \mathbf{K}^{-1} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = [\mathbf{R} \ \mathbf{T}]^{-1} \mathbf{K}^{-1} \mathcal{E} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \quad (13)$$

We solved this equation by robust non-linear least square approach [23] to obtain the rigid transformation $[\mathbf{R} \ \mathbf{T}]$. Please note that we have obtained this rigid transformation over epipolar inliers that are linked with the fundamental matrix $\mathcal{E}$. Based on this estimated rigid transformation and using suitable triangulation [24] we can establish the depth of all matched key-points and generate 3D-point cloud.

2) **DEM Construction by 3D Delaunay Triangulation:** We now construct Digital Elevation Map (DEM) using Delaunay Triangulation (DT) [25] over obtained 3D-point cloud (see figure 6). DT over set of points on a 2D plane fits a surface using 3 points that ensure the circumcircle associated with each triangle contains no other point in its interior. Similarly, 3D-DT [25] fits a surface in 3D space such that no other point in the space is contained inside the circumsphere associated with any tetrahedron.

D. Safe Landing Zone Detection

An appropriate safe zone can be detected by measuring the slope and roughness [1] of the landing site. While slope gives the measure of steepness, roughness indicates the possible presence of hazardous obstacles in the region. Once slope and roughness are determined we segment the “non-hazardous” regions as detailed below.

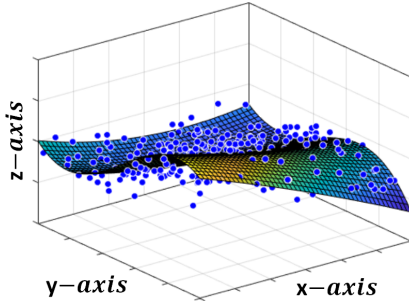


Fig. 2. Smooth underlying surface of DEM obtained over 3D elevation points

1) **Computation of Slope and Roughness Map:** In our problem, we first partition the DEM into several local regions large enough for a helicopter to land. We then fit a 2D quadratic surface (see figure 2) into the set of 3D points belonging to a particular local region based on the following standard quadratic equation:

$$f(x, y) = c_{00} + c_{10}x + c_{01}y + c_{20}x^2 + c_{11}xy + c_{02}y^2 \quad (14)$$

The fitting is achieved by estimating the values of the coefficient using a standard least square approach by minimizing the residual error $l = \sum^n (z - f(x, y))^2$. Once, smooth underlying elevation map is obtained we can compute the slope at a

particular point, say, (x_1, y_1) as $\sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} \Big|_{x_1, y_1}$. The roughness map is computed for the same point (x_1, y_1) as the absolute difference between DEM (z) and the estimated elevation ($\hat{z}_1 = f(x_1, y_1)$) on the smooth underlying map. This slope and roughness computation is performed for all the points in the local DEM to obtain the complete slope and roughness map.

2) **Segmentation of Non-Hazardous Region:** Once the slope and roughness maps are computed, we segment the non-hazardous regions by using the experimentally chosen thresholds over these slope and roughness values. A foreground mask is obtained for the regions below the thresholds for both the maps. An element-wise logical-AND operation is then performed over these two maps to obtain the safe zone(s). If a frame has multiple blob/foreground regions as safe zones we select the one with largest region. Finally, a grassfire (distance) transform is applied over this final map to get the coordinate of the best landing location which is farthest from all hazards.

IV. TIME-COMPLEXITY ANALYSIS

We now present the computational complexity of the epipolar constraint driven MST-based graph clustering. Let n be the number of images/frames in a video. Then, construction of the video representation graph (VRG) would take $O(n^2)$ time. The VRG contains n vertices and n^2 edges. Let us assume the sparse VRG contains m edges. Construction of MST on sparse VRG following Kruskal algorithm takes $O(m \log n)$ time. Further, let the direct neighborhoods of an edge $e_{v_1 v_2}$ connecting the two vertices v_1 and v_2 in this MST be n_1 and n_2 respectively. Then, the inconsistency measure for $e_{v_1 v_2}$ can be computed in $O(n_1) + O(n_2)$ time. So, the inconsistency measure for all m edges in the MST can be computed in $m(O(n_1) + O(n_2))$ time. These m edges need to be sorted based on their inconsistency measure which can be achieved in $O(m \log m)$ time. Finally, $(k - 1)$ most inconsistent edges can be deleted in $(k - 1)O(1)$ time to obtain k clusters. So, the overall complexity of the proposed constrained MST clustering model is: $(O(n^2) + O(m \log n) + m(O(n_1) + O(n_2)) + O(m \log m) + (k - 1)O(1)) \approx O(n^2)$ (as $n_1, n_2, m \ll n$).

V. PROPOSED AHL DATASET

In addition to our proposed framework, this work contributes a dataset especially for developing and testing models for vision guided helicopter landing. As per our knowledge there are no publicly available datasets of such kind and we are the first to propose and render real aerial dataset for such a problem. We have named our dataset as Autonomus Helicopter Landing (AHL) dataset. The dataset was acquired using a drone camera on flight with varying scene and lightning conditions. The dataset is collected in form of image snapshots with a time interval of ≈ 2 sec. All the images are taken at an angle of $\approx 90^\circ$. The images are associated with a geotag having relevant GPS information such as latitude, longitude and altitude position of the drone while auto-capturing images on flight. Such information can

be of importance towards efficient algorithmic development based on scene height and position. Additional camera related information such as Aperture value, Focal length, Shutter-speed, image X and Y resolution as (DPI) etc. are also made available which a developer can utilize to build more accurate models. All information are embedded in the image and can be easily retrieved using suitable commands available in common image processing software (e.g., “*imfinfo*” in Matlab).



Fig. 3. Various scenes of our dataset with and without safe-zones. Frames in first rows are without any safe landing zones while in second row the different safe zones are marked with red-mask.

AHL data consists of 5 sets, namely, $\{Set01, Set02, \dots, Set05\}$, each set correspond to a different scene location (see figure 3). Three out of these five sets i.e., $\{Set02, Set03, Set04\}$ contain safe zones and the rest two are without any safe-zone. The first set mostly consists of images from river banks and nearby congested regions unsuitable for landing. Set02 and Set03 consist of large factory rooftop and a ground near a bridge respectively that can be used for safe helicopter landing. The fourth set consists of a large play field in the vicinity of a small building while the fifth one consists of a mid-sized algae pond (unsafe for landing) that can be mislead as a safe landing zone. We have also provided a ground-truth (GT) annotated mask for frames with safe-zones in the 3 sets. The masks are intended for pixel-level evaluation and localization accuracy of the model. In addition we have also provided frame-level ground-truth annotation indicating whether a frame has a safe-zone or not. The frames with and without safe zones are respectively indicated by flag 1 and flag 0. AHL dataset with all the ground-truth annotations are made available at <https://sites.google.com/site/ivprgroup/helicopter-landing>.

VI. EXPERIMENTAL ANALYSIS

In this section, we first state different evaluation measures used, implementation details and model deployability. This is followed by analysis and comparisons of the proposed clustering model with different clustering techniques. Note that other components of the solution, i.e., natural landmark detection and tracking, digital terrain map construction and safe landing zone detection are kept same.

A. Evaluation measures

As our major contribution in this paper is formulation of a novel clustering approach, we perform a comparison with different clustering techniques using two cluster validation measures, namely, **Dunn Index (DI)** (the higher, the better)

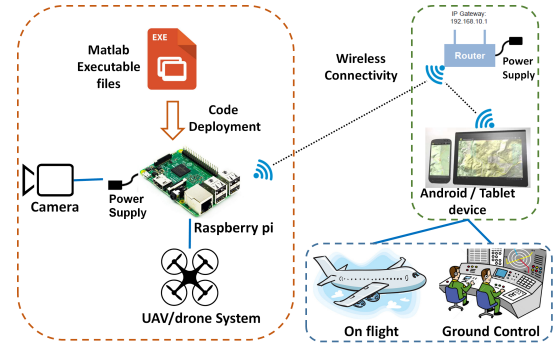


Fig. 4. Deploy mechanism on our drone based system

and **Davies Bouldin Index (DBI)** (the lower the better) [26]. We have also evaluated our method at a frame-level safe-zone detection in comparison with other variants of our pipeline by substituting with various clustering technique. We have used measures like *sensitivity* or *True positive rate (TPR)*, *specificity* or *True negative rate (TNR)*, *precision*, *recall*, *F-measure*, *accuracy* and *confusion matrix* for performance evaluation. We have also shown some qualitative results of pixel-level safe-zone detection and the corresponding ground truth masks.

B. Implementation Details

Our algorithmic pipeline is implemented and developed in Matlab R2018b version running on a 64-bit operating system with 2.6 GHz clock frequency. Our framework has three user parameters, i.e., thresholds for epipolar constraint and those for segmenting slope and roughness maps. We have experimentally set these thresholds to 0.001, 0.1 and 0.02 respectively.

C. Model Deployability

We have deployed and executed our algorithmic framework in a Raspberry Pi based embedded system that can be easily integrated into a UAV/drone as shown in figure 4. The entire process of code deployment is achieved using Matlab 2018b with the aid of some Third-Party Support Packages for Raspberry Pi and Android Devices. The entire communication system can be retrofitted in a Helicopter or a base-station from where the pilot/operator can track safe landing zone(s) on an Android or Tablet Device. We have used User Datagram Protocol (UDP) for establishing connectivity due to its low-latency and multicast nature of transmission.

D. Comparisons based on Cluster Validity Measures

We have chosen standard k-means and hierarchical agglomerative clustering as baseline approaches along with two constrained clustering algorithms, namely, constrained k-means clustering [27] and constrained-spectral-clustering [28] as well as one recently proposed MST based clustering technique, namely, CciMST [29] for comparison. In above four clustering algorithms the optimal number of clusters are determined automatically based on adaptive “Elbow method”. As can be clearly seen from table I, we have achieved the best clustering performance, both in terms of DI and DBI, in all five sets.

TABLE I
CLUSTERING RESULTS FOR DIFFERENT APPROACHES

Video set	Methods	DI($\uparrow$)	DBI($\downarrow$)
Set01	k-means	0.123	0.952
	Agglomerative	0.155	0.781
	COP-Kmeans [27]	0.164	0.744
	CSP [28]	0.181	0.627
	CciMST [29]	1.110	0.576
	Ours	1.280	0.405
Set02	k-means	0.151	1.810
	Agglomerative	0.181	1.760
	COP-Kmeans [27]	0.183	1.300
	CSP [28]	0.226	1.290
	CciMST [29]	0.346	0.924
	Ours	0.570	0.869
Set03	k-means	0.190	1.280
	Agglomerative	0.201	0.934
	COP-Kmeans [27]	0.188	0.871
	CSP [28]	0.210	0.884
	CciMST [29]	0.287	0.713
	Ours	0.307	0.634
Set04	k-means	0.217	1.240
	Agglomerative	0.235	1.160
	COP-Kmeans [27]	0.296	1.140
	CSP [28]	0.422	0.940
	CciMST [29]	0.525	0.649
	Ours	0.853	0.370
Set05	k-means	0.614	0.553
	Agglomerative	0.649	0.518
	COP-Kmeans [27]	0.677	0.525
	CSP [28]	0.714	0.453
	CciMST [29]	0.797	0.438
	Ours	0.867	0.357

We have attained a highest DI value of 1.28 in Set01 and lowest DBI value of 0.375 in Set05. The k-means clustering has performed poorly in all the sets with best attained values of 0.614 and 0.553 in terms of DI and DBI respectively. Agglomerative clustering also followed a similar trend attaining values of 0.649 and 0.781 in terms of DI and DBI respectively. The other two [27], [28] constrained clustering and one recently proposed (CciMST [29]) methods have performed relatively better than the two baseline approaches. CciMST [29] have secured results close to ours in some sets. CciMST [29] have attained best values in terms of DI and DBI as 1.11 and 0.438 which are next best to ours.

TABLE II
FRAME-LEVEL SAFE ZONE COMPARISON WITH VARIOUS COMPETING CLUSTERING METHODS

Methods	Criteria					
	TPR	TNR	Precision	Recall	F-score	Accuracy
k-means	0.125	0.952	0.438	0.125	0.194	0.761
Agglomerative	0.196	0.947	0.524	0.196	0.286	0.774
COP k-means [27]	0.250	0.952	0.609	0.250	0.354	0.790
CSP [28]	0.304	0.925	0.548	0.304	0.391	0.782
CciMST [29]	0.339	0.941	0.633	0.339	0.442	0.802
Our Proposed	0.554	0.952	0.775	0.554	0.646	0.860

E. Comparisons for Frame Level Safe-zone Detection

We have compared frame-level safe-zone detection performances by substituting the above clustering techniques [27]–[29] in our pipeline to show the effectiveness of our formulation. In this regard, we have presented confusion matrices

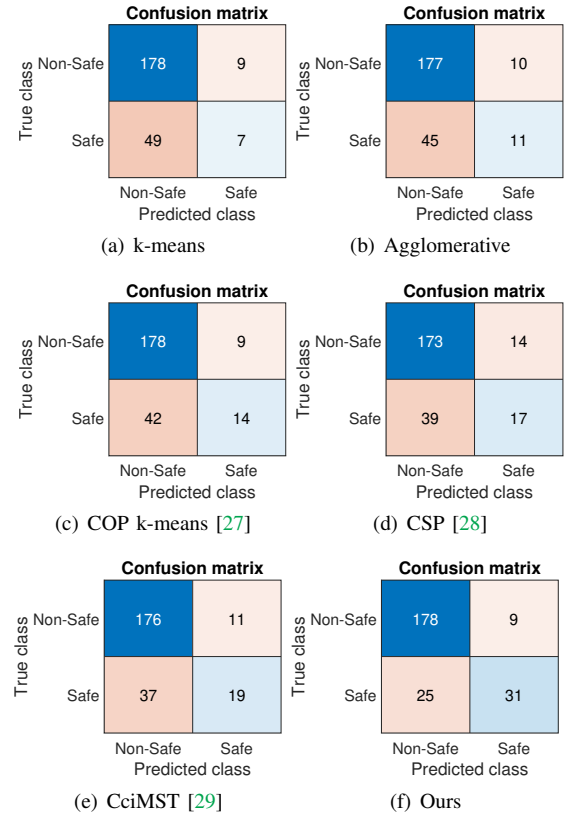


Fig. 5. Confusion Matrices for various clustering methods for Frame-level Safe-zone detection

(CMs) (see figure 5) to better visualize the frame-level detection performance for these different competing methods. It is clearly evident that with k-means and agglomerative clustering the frame-level detection performance are much inferior resulting in lower sensitivity (TPR) of 0.125 and 0.196 respectively (see table II). This is also in agreement with explicit clustering results in terms of DBI and DI. Our proposed solution has in fact outperformed all five competing clustering strategies yielding best sensitivity (TPR) and specificity (TNR) values of 0.554 and 0.952 respectively. We have also attained an F-score of 0.646 whereas the 2nd best performing method CciMST [29] considerably lags behind with a F-score of 0.442. We have also shown some typical pixel-level detection performance (see figure 6) along with 3D point cloud and DEM as intermediate results. The above performance analysis clearly indicate the superiority of our method in terms of safe-zone detection. Please note that our proposed solution being an unsupervised approach requires no training. We simply mark frames with non-hazardous regions as safe which are then compared with GT annotation to compute the CMs (see figure 5).

VII. CONCLUSION

In this paper, we have presented a solution to the safe zone detection for helicopter landing from videos captured by a drone in a completely unconstrained manner. An epipolar constraint driven MST based clustering algorithm is designed as a part of the solution. A new (AHL) dataset consisting of five sets of aerial videos has publicly been made available.

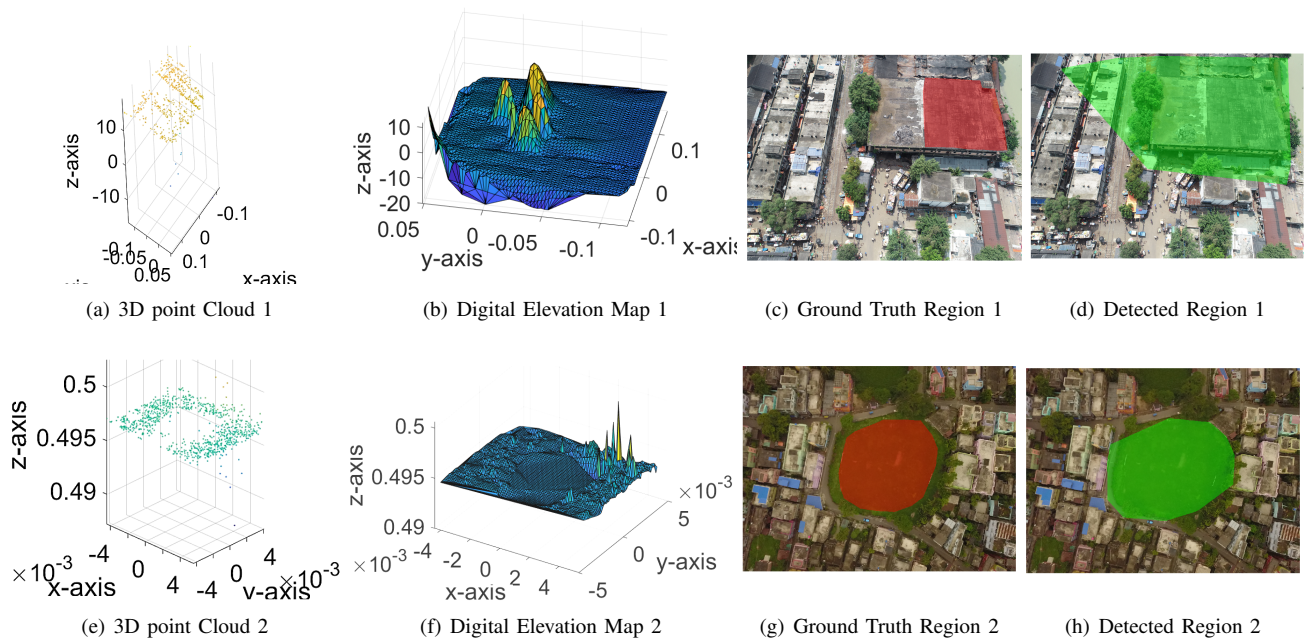


Fig. 6. 3D Point Cloud, Digital Elevation Map, Ground Truth and Detected safe-zone in a frame from Set02 (top row) and Set04 (bottom row).

Experiments clearly show the effectiveness of proposed formulation. In future, we will primarily focus on the pixel level safe zone detection by improving camera calibration.

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